

title: H · Data Migration Plan description: How to move 1 M+ employee records from legacy PIS / spreadsheets / paper 201 files into the new HRIS without repeating the Novopay disaster — sources, target model, quality profile, waves, reconciliation, cutover, rollback.

H · Data Migration Plan

!!! danger “Why this paper exists” Every large government HRIS that has failed in the past 15 years — **Novopay (NZ 2012), Queensland Health payroll (AU 2010), UK NHS NPfIT (2011)** — failed on the same axis: **the legacy data was worse than anyone admitted, and the cutover was rushed**. This paper treats data migration as a **first-class workstream** with its own owner, its own budget line (accounted for in Paper F), and its own go / no-go gate before payroll parallel-run at M6.

H.1 Guiding principles

1. **Migration is not an event. It is a system.** Sources continue to change during the 12-month build. The migration pipeline must run continuously from M3 onwards, not as a one-shot event at M7.
2. **No employee is migrated unless four gates pass.** Identity resolved, employment history reconstructable, mandatory fields non-null, no active disciplinary lock. Records that fail any gate go to a **quarantine queue** for HR review, never silently dropped.
3. **The legacy system stays authoritative until M6.** New HRIS is a read-replica through M5. Only after payroll parallel-run passes at M6 does the new system become authoritative for its bounded contexts.
4. **Every migrated record is traceable to a legacy source row.** A `legacy_source_ref` column exists on every table with a foreign origin. Auditability is not optional.
5. **Waves before big bang.** Migration proceeds region by region — 3 pilot SDOs first, then batched by cluster of similar SDOs. No single all-Philippines cutover.

H.2 Sources — what we are actually pulling from

The PBD is silent on legacy sources beyond naming the incumbent PIS. Reality across DepEd is that employee data lives in a dozen systems of record, some of which are spreadsheets on personal drives.

Source	Type	Estimated records	Data quality baseline	Owner
PIS (legacy Personnel Information System)	Oracle DB, ~2007 vintage	~ 900,000 employees	Moderate — canonical for teaching + related-teaching; ghost records common	ICTS-SDD
PSIPOP (Plantilla / positions)	DBM-owned Postgres	~ 1,050,000 items	High — authoritative for plantilla items	DBM
CSC-BEA position master	CSC-owned	~ 1,050,000 refs	High — but change lag vs PSIPOP	CSC
Payroll runs, past 24 months	Flat files + Excel	~ 26 M line items	Variable — used to seed anomaly detector	DepEd Finance
RSP (Recruitment Selection Placement) archives	SharePoint + shared drives	~ 180,000 postings	Low — inconsistent formats, PDFs, scans	HR (per RO)
201 file scans (paper)	Physical + partial PDFs	~ 850,000 folders	Very low — the manual-entry backstop	HR (per school)
Time & attendance (biometric exports)	Vendor-specific CSVs	~ 340 M records/yr	Variable	Per SDO
GSIS / Pag-IBIG / PhilHealth / SSS / BIR	External regulators	~ 900,000 refs each	High — regulators are authoritative	External
Leave ledgers	Excel per RO	~ 900,000 balances	Low — reconciliation gaps common	HR (per RO)
PDS files (CSC standard form)	Word / PDF / partial DB	~ 900,000	Moderate — self-declared, needs eKYC dedup	Employee-owned

Total unique canonical records to reconcile: ~ 1.0 M–1.1 M employees.

H.3 Target model — what we are migrating into

Target is the PostgreSQL 16 schema defined in [Paper C](#) — three bounded contexts:

- **Core HR** — `employee`, `employment_history`, `plantilla_item`, `assignment`, `education`, `eligibility`, `training`, `leave_ledger`, `document`
- **Payroll** — populated only during the M6 parallel-run, not before
- **Recruitment** — historical postings and shortlists rebuilt from RSP archives

Every target table has:

- `legacy_source_ref` (jsonb) — pointer(s) to source rows
- `migration_wave` (int) — which cutover batch
- `migration_confidence` (enum: `verified`, `derived`, `stub`) — how much of this row we trust
- `data_quality_flags` (int[]) — bitmask of issues (see H.5)

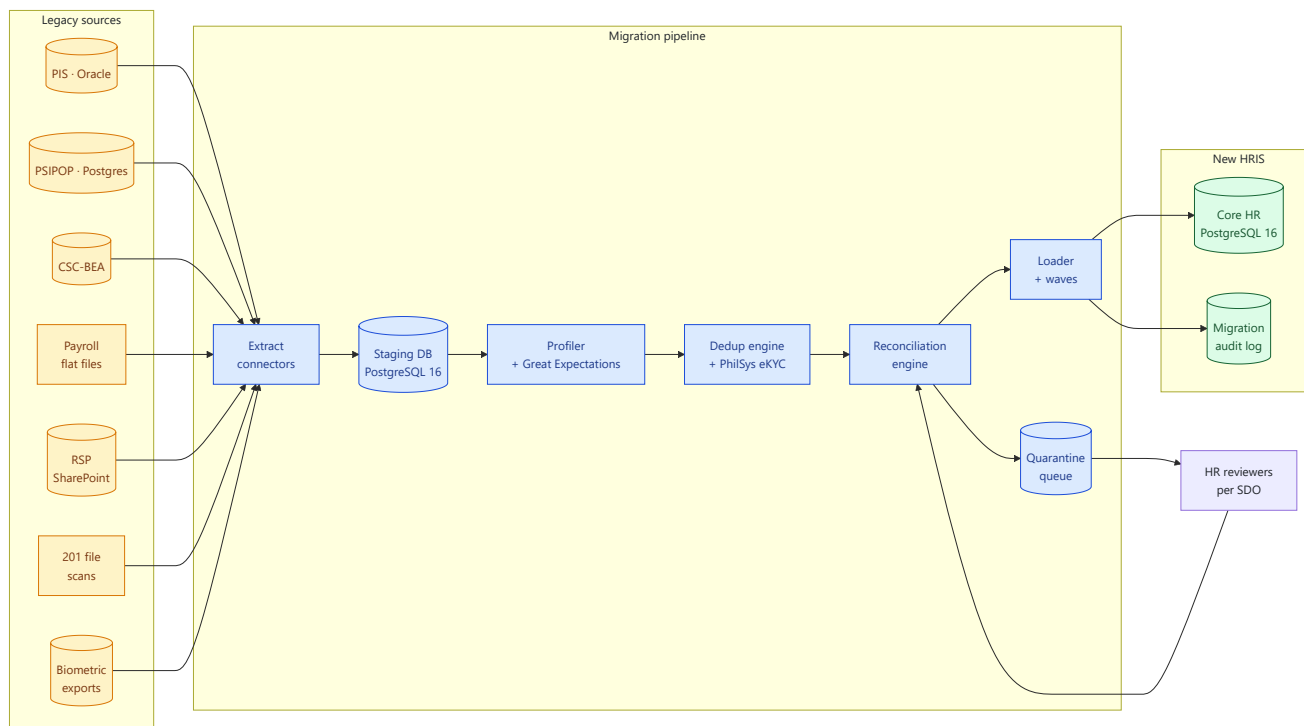
H.4 Expected data-quality issues (baseline profile)

Based on public audits of PIS (COA 2019, 2022, 2024) and adjacent public-sector HRIS profiling:

Issue	Estimated affected records	Handling
Ghost employees (in PIS, not in PSIPOP or regulator files)	3,000 – 8,000	Quarantine → HR verification → written off with audit trail
Duplicate employees (same person, multiple PIS IDs)	12,000 – 25,000	PhilSys eKYC-driven merge (D \$13); manual review for non-match
Missing PhilSys number	~ 40 % initially	Progressive enrichment via ESS on first login
Missing / stale civil-status	~ 15 %	Employee self-service update, deadline enforced
Payroll and PIS employee-ID mismatch	~ 8 %	ID reconciliation crosswalk table
Legacy 8-bit encoding on names	Widespread	UTF-8 normalisation with ñ / ligature preservation
Invalid dates (birth date > hire date, hire before agency existed)	~ 2 %	Quarantine + HR audit
Position codes not in PSIPOP master	~ 3 %	Escalate to DBM; provisional plantilla_item stub
Assignment history discontinuities	~ 12 %	Reconstruct from payroll runs; flag if unrecoverable
Multiple concurrent active assignments	~ 1.5 %	Business-rule violation; HR must resolve pre-cutover
Leave-balance disputes	~ 6 %	Freeze legacy balance at cutover; disputes resolved in new system
Documents in wrong 201 folder	Unknowable	Reindex during scanning wave; ML-assisted OCR

Every issue produces a **data-quality report** consumed by the HR governance team weekly during migration.

H.5 Migration architecture



Components:

- **Extract connectors** — Airbyte or custom Python + `psycopg` for Oracle; `pypdf` + Tesseract OCR for scanned 201 files (with human-in-the-loop review).
- **Staging DB** — a dedicated PostgreSQL 16 instance mirroring the target schema plus source-tracing columns. All transformations happen here; production is never touched by raw source data.
- **Profiler** — Great Expectations + custom dbt tests. Runs daily against staging; publishes a quality dashboard.
- **Dedup engine** — deterministic PhilSys eKYC match first (highest confidence), then fuzzy match on (birthdate, mother's maiden name, first-hire date) with a 0.85 similarity threshold; ambiguous cases go to quarantine.
- **Reconciliation engine** — cross-checks staging against PSIPOP (positions), regulator files (identity), and payroll history (existence-of-life). Emits reconciliation reports per SDO.
- **Loader** — batches records into waves, writes to production, updates `migration_wave` + `migration_confidence` + `data_quality_flags`.

H.6 Wave plan — regional rollout, not big bang

M3 • Pilot (3 SDOs, ~20,000 employees) — one urban (NCR), one rural (Cordillera), one geographically difficult (MIMAROPA). Purpose: validate the pipeline and quality gates on a manageable population.

M4 • Expansion 1 (5 regions, ~250,000 employees) — Regions with mature ICT capacity. Purpose: catch edge cases the pilot missed.

M5 • Expansion 2 (7 regions, ~450,000 employees) — Includes ARMM/BARMM special considerations (bilingual, multi-currency for hazard pay).

M6 • Expansion 3 (remaining regions, ~280,000 employees) — Final wave. All regions live in the new HRIS as a **read-replica** — legacy PIS still authoritative for payroll.

M6 · Payroll parallel-run gate — three consecutive payroll cycles matched to legacy within tolerance (Paper F §F.11.1). Only after this gate does the new system become **authoritative for payroll**.

M7 · Cutover — legacy PIS moves to read-only. New HRIS is single source of truth.

M8+ · Long tail — quarantine backlog worked down over 90 days; unrecoverable records written off with COA-visible audit trail.

H.7 Reconciliation reports (published weekly)

Report	Consumers	Fail action
Employee count delta (staging vs source vs target)	ICTS-SDD, HR governance	Investigate any variance > 0.1 %
Payroll-total variance (parallel-run cycles)	Finance, Payroll officers	Fail M6 gate if variance > 0.5 %
Regulator remittance reconciliation (GSIS / Pag-IBIG / PhilHealth / SSS / BIR)	Finance + regulators	Fail M6 gate if variance > 0.05 % on any regulator
Quarantine queue depth	HR governance	Escalate to Program Manager if depth grows week-on-week
Data quality flag distribution	Migration Lead, HR	Trigger remediation sprint if any flag > 10 %
Dedup confidence histogram	Data Engineer, HR	Manual review batch if fuzzy-match rate > 5 %
SDO adoption metrics (percentage of records migrated per SDO)	Program Manager	Regional escalation path

H.8 Rollback and freeze windows

Two rollback surfaces:

1. **Per-wave rollback (M3–M6)** — waves are transaction-scoped. Any failed wave can be re-run without affecting completed waves. Loader keeps a `pre_load_snapshot` for every wave for 90 days.
2. **Post-cutover rollback (M7 onwards)** — if a P1 defect is detected in the first 72 h post-cutover, the system can revert to **legacy PIS as authoritative for reads**, with all new-HRIS writes replayed against legacy through a dedicated reverse-sync. This is the “**nuclear option**” — never invoked lightly, but pre-designed so it can be invoked at all. Requires a joint Program Manager + DepEd CIO + Finance Head signoff.

Freeze windows:

- **72 h before each wave** — legacy source systems freeze writes on the affected records. HR officers pre-notified 14 days in advance.
- **7 d before M6 gate** — all payroll changes freeze; legacy runs one final cycle for parallel-run baseline.
- **48 h at M7 cutover** — full write-freeze across the whole system.

H.9 Legacy sunset

Legacy PIS is **not** decommissioned at M8. It remains available in **read-only, air-gapped mode for 24 months** post-cutover, then archived to cold storage per RA 9470 (National Archives Act) retention. This gives auditors and COA field officers a fallback for historical queries and preserves any legacy audit trail the new system may not have absorbed.

Sunset schedule:

- **M8** — legacy set to read-only
- **M8 + 12 mo** — legacy moved to isolated network segment; access via ticket only
- **M8 + 24 mo** — legacy backed up to cold storage; database instance decommissioned
- **M8 + 10 yr** — cold storage disposed per COA / National Archives retention rules

H.10 Ownership, cost, and cross-references

Aspect	Details
Workstream owner	Migration Lead + 2 Migration Specialists (see Paper F §F.3.2)
Duration	M3 onwards, 8 months active + 3 months quarantine remediation
Budget	Included in Paper F personnel (Migration Specialist × 2) + infra (Staging DB in non-prod) + software (Great Expectations OSS, Airbyte OSS, OCR license ~ PHP 400 K)
Risk register anchor	R1 (Payroll cutover data loss) and R3 (Ghost / duplicate employees) in Paper B §15
Value-add tie-ins	D §4 (Payroll anomaly detector consumes migration data), D §13 (PhilSys eKYC dedup)
Compliance anchor	RA 10173 processing must be documented in Paper I — migration is a covered processing activity

H.11 Success criteria

The migration is declared successful when **all** of the following hold at M8:

1. $\geq 99.5\%$ of expected employees are in the new HRIS with `migration_confidence` of `verified` or `derived`.
2. Quarantine queue < 5,000 records, all with owner assigned.
3. Three consecutive payroll cycles reconciled to legacy within tolerance.
4. All nine regulator remittances reconciled within 0.05 %.
5. COA has been given walk-through access to the migration audit log.
6. Data Privacy Officer has signed off on the migration under RA 10173 (see [Paper I](#)).
7. Written sign-off from ICTS-SDD, DepEd CIO, and Finance Head.

Anything less is not a cutover — it is a rollout with known defects, and it must be documented as such before M8 payment is claimed.